

Concept of Network Policy Server (NPS)

What is Network Policy Server (NPS)?

Network Policy Server (NPS) is a role in Windows Server that acts as a **RADIUS (Remote Authentication Dial-In User Service) server and proxy**. It is used to centralize the authentication, authorization, and accounting of network access requests, providing secure access control for remote clients and network devices.

NPS is a critical component in managing network access policies for VPNs, wireless networks, dial-up connections, and wired 802.1X authentication.

Key Functions of NPS

1.

RADIUS Server:

NPS receives connection requests from RADIUS clients such as VPN servers, wireless access points, or switches, authenticates users or devices, and authorizes or denies access based on policies.

2.

RADIUS Proxy:

NPS can forward RADIUS requests to other RADIUS servers when acting as a proxy, useful for large or distributed networks.

3.
Network Access Protection (NAP) Server:
Integrates with NAP to enforce health policies on client devices before granting access.
-

Components of NPS

- **RADIUS Clients:** Network access servers that send authentication requests to NPS (e.g., VPN servers, wireless access points).
 - **Connection Request Policies:** Rules that determine how NPS processes connection requests.
 - **Network Policies:** Conditions and settings that determine who can connect, when, and under what conditions.
 - **Accounting:** Tracks usage data for connections to help with auditing and billing.
-

How NPS Works

When a user or device attempts to connect to a network resource, the access server (e.g., VPN, wireless AP) sends a RADIUS request to NPS. NPS then authenticates the credentials (e.g., via Active Directory), checks policies, and responds to allow or deny access. It can also log accounting information about the session.

Why Use NPS?

- Centralized authentication and authorization for network access.
- Policy-based access control for enhanced security.
- Support for multiple connection types (VPN, wireless, wired 802.1X).
- Integration with Active Directory for user and group-based policies.
- Enables auditing and accounting of network usage.

Installing and Managing NPS

- Install the NPS role via Server Manager.
- Configure RADIUS clients (network devices) that send requests.
- Define Connection Request Policies and Network Policies via the NPS console.
- Enable accounting if needed.
- Use Event Viewer and logs to monitor NPS activity.